

E-Commerce Sales & Customer Performance Analysis

Data Analyst Project | Python • SQL • Power BI

Project Report

This project analyzes 5,000 e-commerce orders to understand sales performance, customer behavior, product categories, regional performance, payment methods, delivery performance, and time-based revenue trends. Python was used for initial data preparation, SQL for analytical querying, and Power BI for interactive reporting.

Total Revenue	5.11M
Total Orders	5,000
Total Customers	989
Average Order Value	1.02K
Average Customer Rating	2.97
Average Delivery Time	6.12 days

1. Project Overview

The objective is to transform raw e-commerce order data into actionable business insights. The analysis focuses on product performance, regional sales, customer concentration, payment behavior, monthly revenue, and operational performance.

Business Questions

- Which product category generates the highest and lowest revenue?
- Which category has the highest quantity sold and average order value?
- Which region generates the highest revenue?
- Which region has the fastest average delivery time and highest customer rating?
- Which payment method contributes the highest transaction value?
- Which month records the highest and lowest revenue?
- Who are the top customers by total spending?
- What percentage of customers are repeat customers?
- What is the overall customer rating and delivery performance?

Technology Stack

Python (Pandas) • MySQL/SQL • Power BI • Data Cleaning • Exploratory Analysis • Aggregations • Joins • CTEs • Window Functions • Dashboarding

2. Data & Data Preparation

The project uses an e-commerce sales and customers dataset containing 5,000 records and 989 unique customers. The available order dates range from 01 January 2022 to 09 September 2035.

Python Data Preparation Workflow

- Loaded the raw CSV file using Pandas.
- Checked missing values.
- Standardized column names by converting them to lowercase and replacing spaces with underscores.
- Converted the mixed-format order_date field into a consistent datetime format.
- Exported the cleaned dataset as ecommerce_sales_customers.csv for downstream analysis.

Python Data Preprocessing Script (data_cleaning.py):

```
import pandas as pd

# Ingest raw dataset
df = pd.read_csv('ecommerce.csv')

# Verify null values
print(df.isnull().sum())

# Standardize column headers to lowercase snake_case
df.columns = df.columns.str.lower().str.replace(' ', '_')

# Standardize mixed datetime formats to ISO standard
df["order_date"] = pd.to_datetime(df["order_date"], format="mixed")
df["order_date"] = df["order_date"].dt.strftime("%Y-%m-%d")

# Export standardized dataset for MySQL and Power BI
df.to_csv("ecommerce_sales_customers.csv", index=False)
```

Data Quality Notes

The raw order_date field contains mixed date formats, so the Python cleaning step standardizes the date before SQL and Power BI analysis. The cleaned file should be used as the main BI/SQL input while retaining the original CSV as raw data.

3. SQL Analysis

SQL was used as the analytical layer to answer business questions through filtering, aggregation, GROUP BY, HAVING, joins, CTEs, and window functions.

SQL Questions

Section 1: Sales & Business Performance

1. Which product category generates the highest and lowest revenue?

```
(SELECT product_category, SUM(revenue) AS total_revenue
FROM details
GROUP BY product_category
ORDER BY total_revenue DESC
LIMIT 1)
```

UNION ALL

```
(SELECT product_category, SUM(revenue) AS total_revenue
FROM details
GROUP BY product_category
ORDER BY total_revenue ASC
LIMIT 1
);
```

product_category	total_revenue
Electronics	1829899.22
Beauty	765860.88

2. Which category has the highest quantity sold and average order value?

```
SELECT product_category, SUM(quantity) AS total_quantity,
        ROUND(AVG(revenue),2) AS avg_order_value
FROM details
GROUP BY product_category
ORDER BY total_quantity DESC, avg_order_value DESC
LIMIT 1;
```

product_category	total_quantity	avg_order_value
Electronics	7109	1029.77

3. Which region generates the highest revenue?

```
SELECT region, SUM(revenue) AS total_revenue
FROM details
GROUP BY region
ORDER BY total_revenue DESC
LIMIT 1;
```

region	total_revenue
West	1345582.16

4. Which region has the fastest average delivery time?

```
SELECT region, ROUND(AVG(delivery_days),2) AS fastest_delivery
FROM details
GROUP BY region
ORDER BY fastest_delivery ASC
LIMIT 1;
```

region	fastest_delivery
West	6.08

5. Which region has the highest customer rating?

```
SELECT region, ROUND(AVG(customer_rating),2) AS highest_rating
FROM details
GROUP BY region
ORDER BY highest_rating DESC
LIMIT 1;
```

region	highest_rating
North	2.99

Section 2: Customers & Transactions

1. Which payment method contributes the highest transaction value?

```
SELECT payment_method, SUM(revenue) AS total_transaction_value
FROM details
GROUP BY payment_method
ORDER BY total_transaction_value DESC
LIMIT 1;
```

payment_method	total_transaction_value
Card	2366248.20

2. Which month records the highest and lowest revenue?

```
(SELECT DATE_FORMAT(order_date, '%Y-%m') AS month,
        SUM(revenue) AS total_revenue
FROM details
GROUP BY DATE_FORMAT(order_date, '%Y-%m')
ORDER BY total_revenue DESC
LIMIT 1)
```

UNION ALL

```
(SELECT DATE_FORMAT(order_date, '%Y-%m') AS month,
        SUM(revenue) AS total_revenue
FROM details
GROUP BY DATE_FORMAT(order_date, '%Y-%m')
ORDER BY total_revenue ASC
LIMIT 1);
```

month	total_revenue
2026-08	44728.82
2035-09	4952.23

3. Who are the top 10 customers by total spending with their ties?

```
WITH customer_rank AS
(SELECT customer_id,
        SUM(revenue) as total_revenue,
        DENSE_RANK() OVER(ORDER BY SUM(revenue) DESC) AS dr
FROM details
GROUP BY customer_id)
SELECT customer_id, total_revenue
FROM customer_rank
WHERE dr <= 10;
```

customer_id	total_revenue
1663	17680.69
1955	16265.15
1675	15036.46
1276	15023.71
1647	14894.97
1804	14330.98
1193	14151.65
1836	14110.78
1957	13956.95
1735	13146.25

4. What percentage of customers are repeat customers?

```
WITH count_orders AS
  (SELECT customer_id,
         COUNT(*) AS total_orders
   FROM details
  GROUP BY customer_id
 )
SELECT ROUND(COUNT(CASE WHEN total_orders > 1 THEN 1 END) * 100.0 /
             COUNT(*),2) AS repeat_customer_percentage
FROM count_orders;
```

repeat_customer_percentage
96.06

5. What is the overall customer rating and average delivery time?

```
SELECT ROUND(AVG(customer_rating),2) AS overall_rating,
       ROUND(AVG(delivery_days),2) AS avg_delivery_time
FROM details;
```

overall_rating	avg_delivery_time
2.97	6.12

4. Sales & Product Analysis

Category-level analysis shows that Electronics is the strongest revenue contributor, followed by Clothing, Home, and Beauty.

Category	Revenue	Revenue Share	Quantity	Avg. Order Value
Electronics	1.83M	35.81%	7,109	1.03K
Clothing	1.53M	29.98%	6,171	1.00K
Home	982.08K	19.22%	3,949	1.01K
Beauty	765.86K	14.99%	2,995	1.06K

Key Findings

- Electronics leads revenue with 1.83M.
- Electronics also has the highest quantity sold with approximately 7,109 units.
- Beauty has the highest average order value at approximately 1.06K.
- The combination of low Beauty revenue and high average order value suggests that increasing order frequency could be a useful growth opportunity.

5. Regional Analysis

Regional analysis compares revenue, order volume, customer ratings, and delivery performance.

Region	Revenue	Share	Orders	Avg. Rating	Avg. Delivery
West	1.35M	26.33%	1,289	2.96	6.08
North	1.28M	25.08%	1,276	2.99	6.12
South	1.25M	24.40%	1,208	2.97	6.17
East	1.24M	24.19%	1,227	2.97	6.10

- West is the highest-revenue region at approximately 1.35M.
- West also has the fastest average delivery at approximately 6.08 days.
- North has the highest average customer rating at approximately 2.99.

6. Time & Payment Analysis

Payment Method

Payment Method	Revenue	Revenue Share
Card	2.37M	46.31%
COD	1.79M	35.00%
Wallet	955.12K	18.69%

Monthly Revenue

Month	Revenue
January	426.37K
February	413.89K
March	425.05K
April	398.72K
May	479.77K
June	442.94K
July	440.57K
August	460.87K

September	402.37K
October	407.87K
November	410.66K
December	400.69K

- May records the highest revenue at approximately 479.77K.
- April records the lowest revenue at approximately 398.72K.

7. Customer Analysis

The dataset contains 989 unique customers. Based on customer order frequency, approximately 96.06% of customers placed more than one order.

Top 10 Customers by Total Spending

Rank	Customer ID	Total Revenue
1	1663	17,680.69
2	1955	16,265.15
3	1675	15,036.46
4	1276	15,023.71
5	1647	14,894.97
6	1804	14,330.98
7	1193	14,151.65
8	1836	14,110.78
9	1957	13,956.95
10	1735	13,146.25

Customer 1663 is the highest-spending customer with approximately 17,680.69 in total revenue. The top 10 customers together generated 148,597.59.

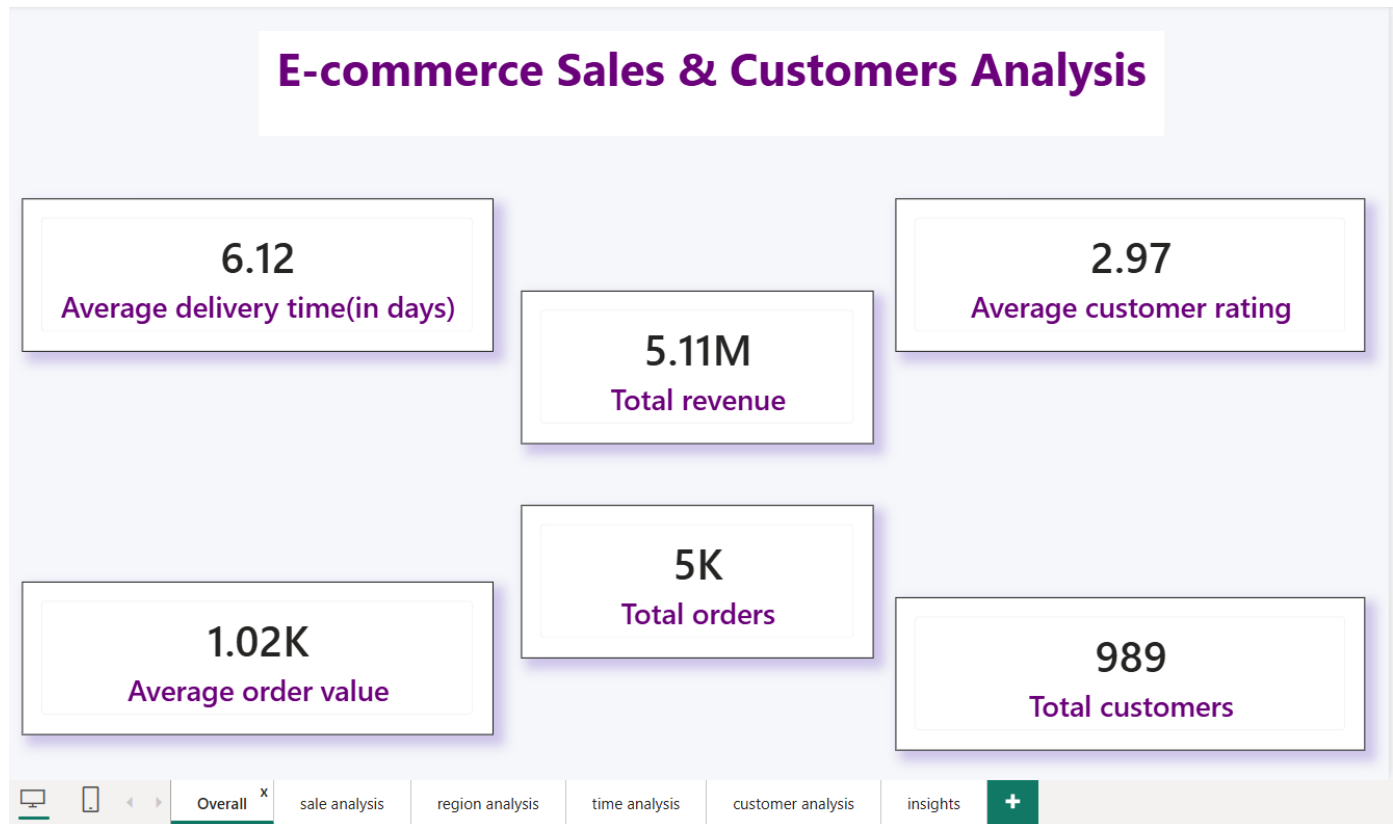
Customer Satisfaction & Operations

- Overall average customer rating is 2.97.
- Overall average delivery time is 6.12 days.
- The rating below 3.0 indicates customer satisfaction should be treated as an improvement area.
- Regional delivery and rating differences should be investigated together to understand operational and customer-experience drivers.

8. Power BI Dashboard

The Power BI report is organized into Overall, Sales Analysis, Region Analysis, Time Analysis, Customer Analysis, and Insights pages. The dashboard combines KPI cards, bar charts, a monthly trend chart, payment visuals, slicers, customer rankings, and business recommendations.

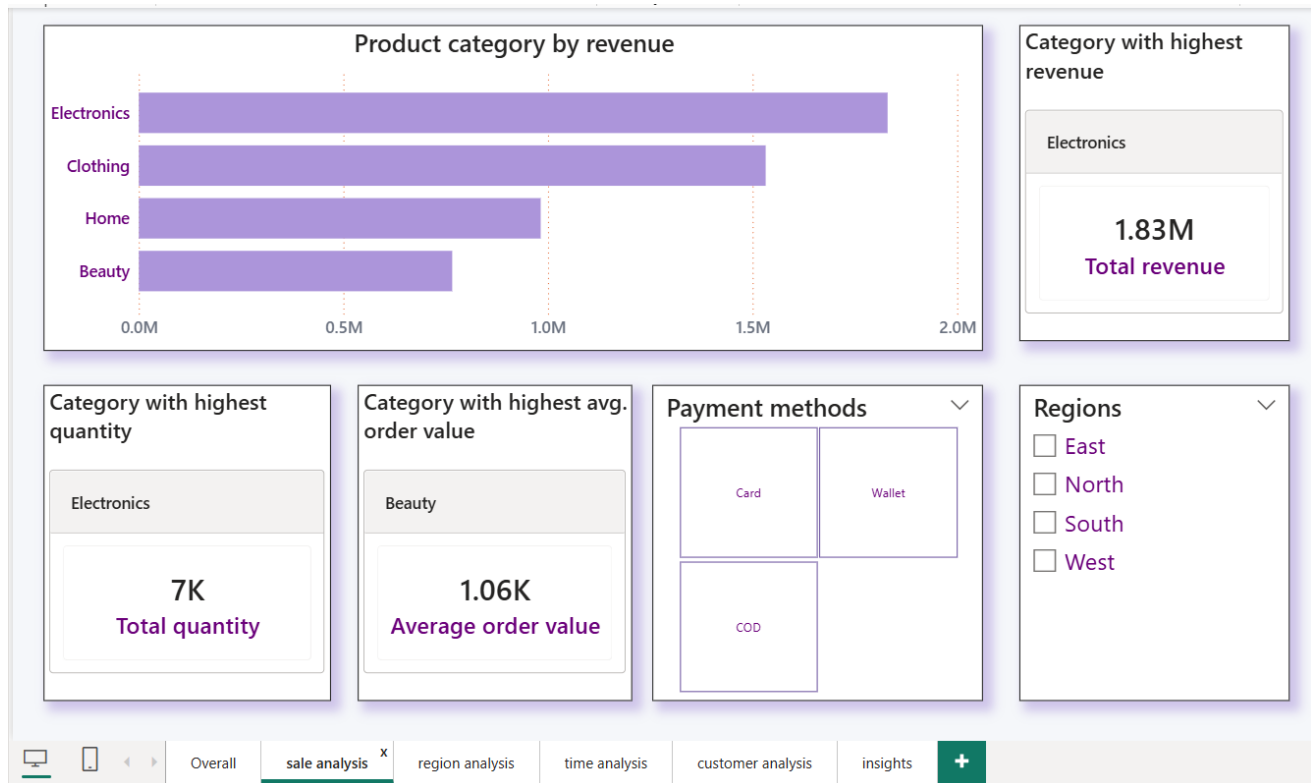
Dashboard Page 1 — Overall Performance



Key Findings:

- The business generated 5.11M in revenue from 5,000 orders across 989 customers. With an average order value of 1.02K, the main improvement area is customer experience, reflected in the 2.97 average rating and 6.12-day delivery time.
- Average delivery time is 6.12 days.
- Average customer rating is 2.97, making customer experience a key improvement area.
- Average order value is 1.02K.

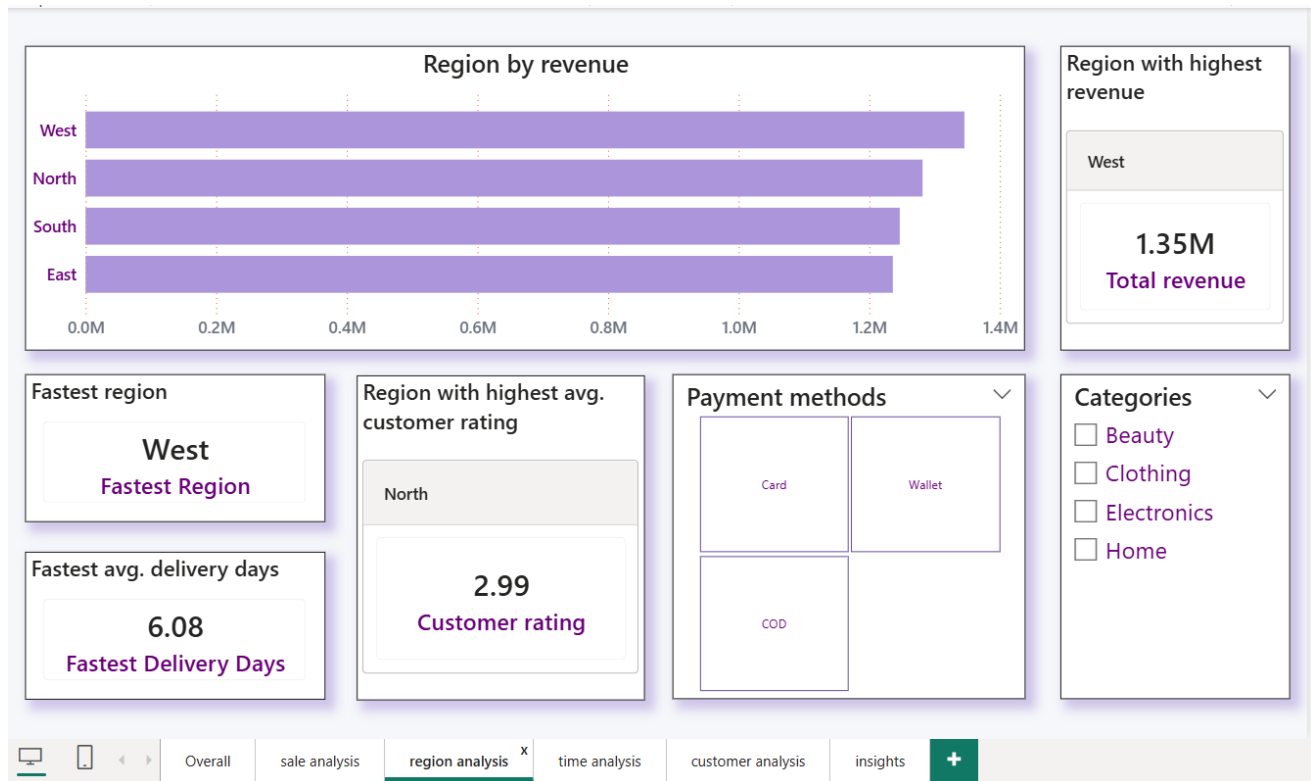
Dashboard Page 2 — Sales Analysis



Key Finding:

- Electronics leads both revenue (1.83M) and quantity sold (7,109 units), making it the strongest category.
- Beauty has the lowest revenue (765.86K) but the highest average order value (1.06K), suggesting an opportunity to grow revenue by increasing order frequency.

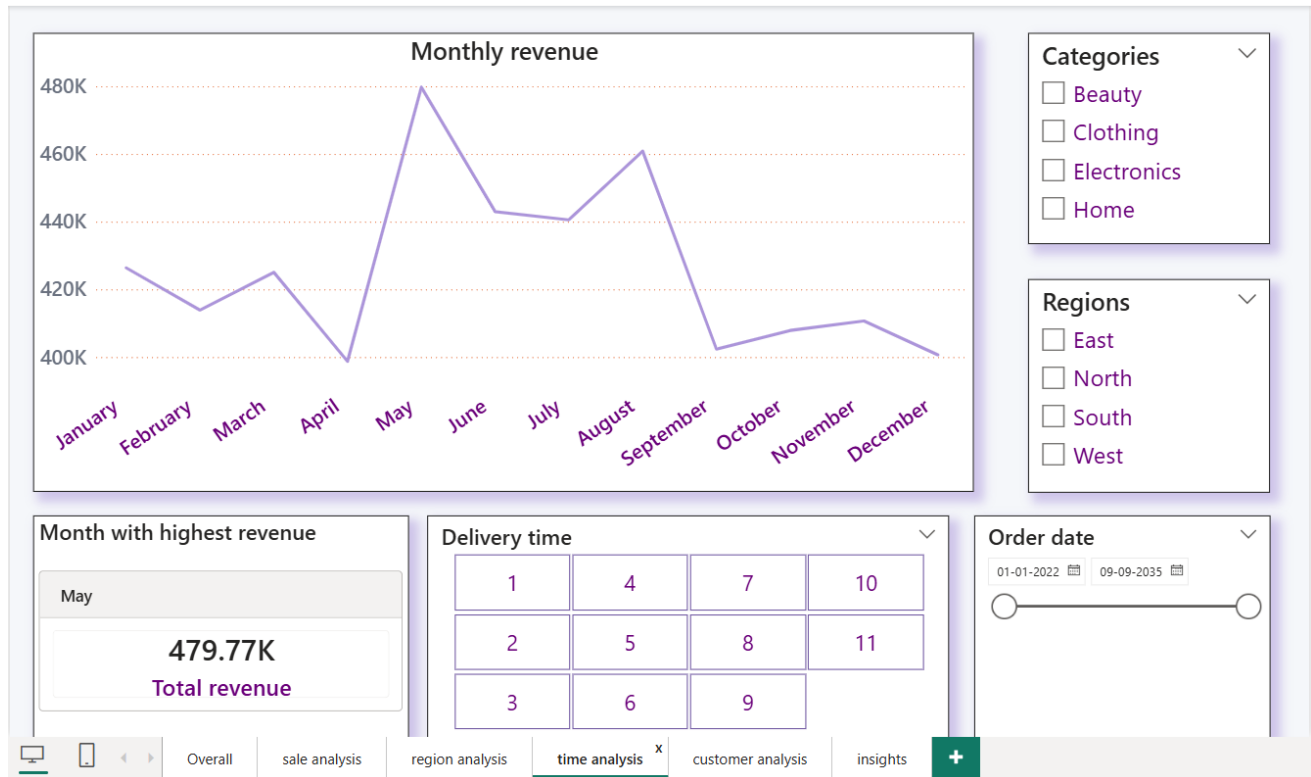
Dashboard Page 3 — Regional Analysis



Key Finding:

- West leads regional revenue at 1.35M and has the fastest average delivery time at 6.08 days.
- North has the highest average rating at 2.99.
- West can be studied for operational efficiency.
- North can provide clues for customer experience.

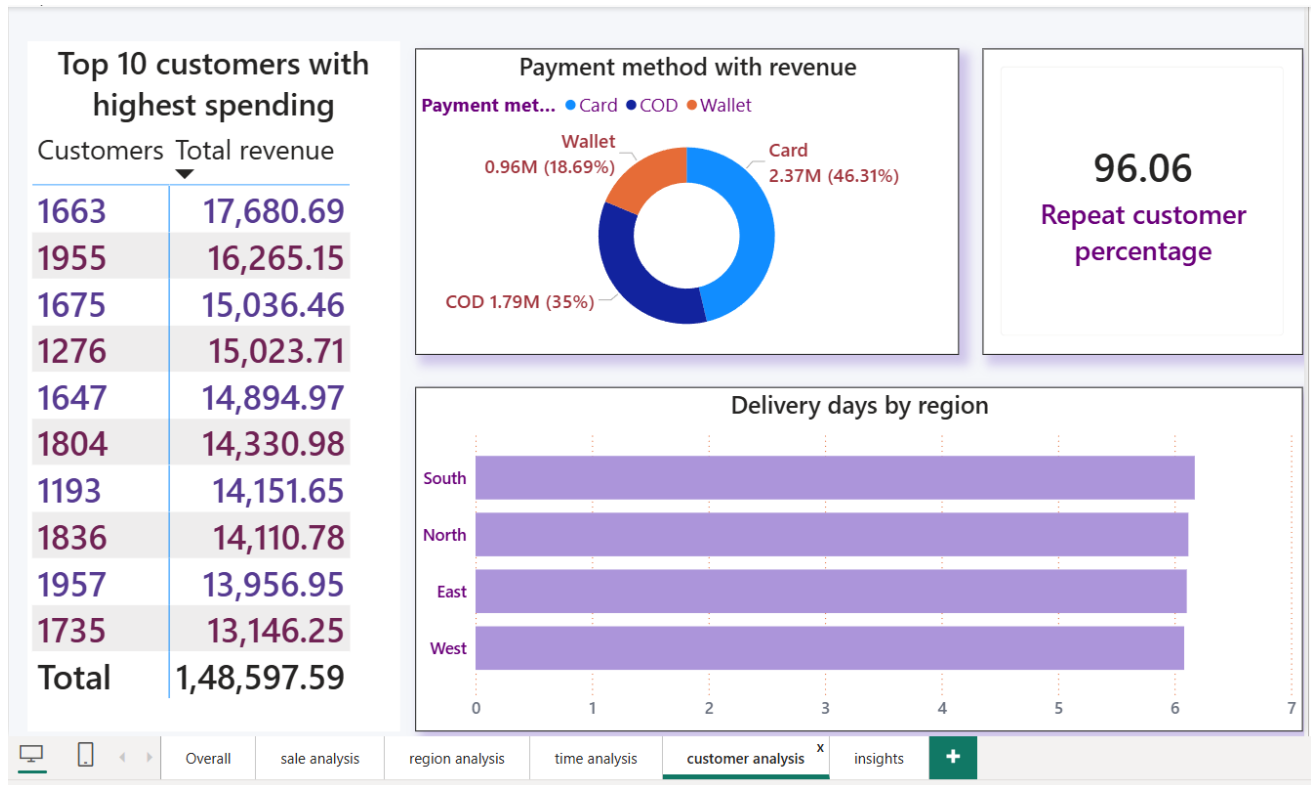
Dashboard Page 4 — Time Analysis



Key Finding:

- May records the highest revenue at 479.77K.
- April records the lowest at 398.72K.
- These month-of-year patterns can support campaign and inventory planning, but seasonality should be validated with year-over-year analysis.

Dashboard Page 5 — Customer Analysis

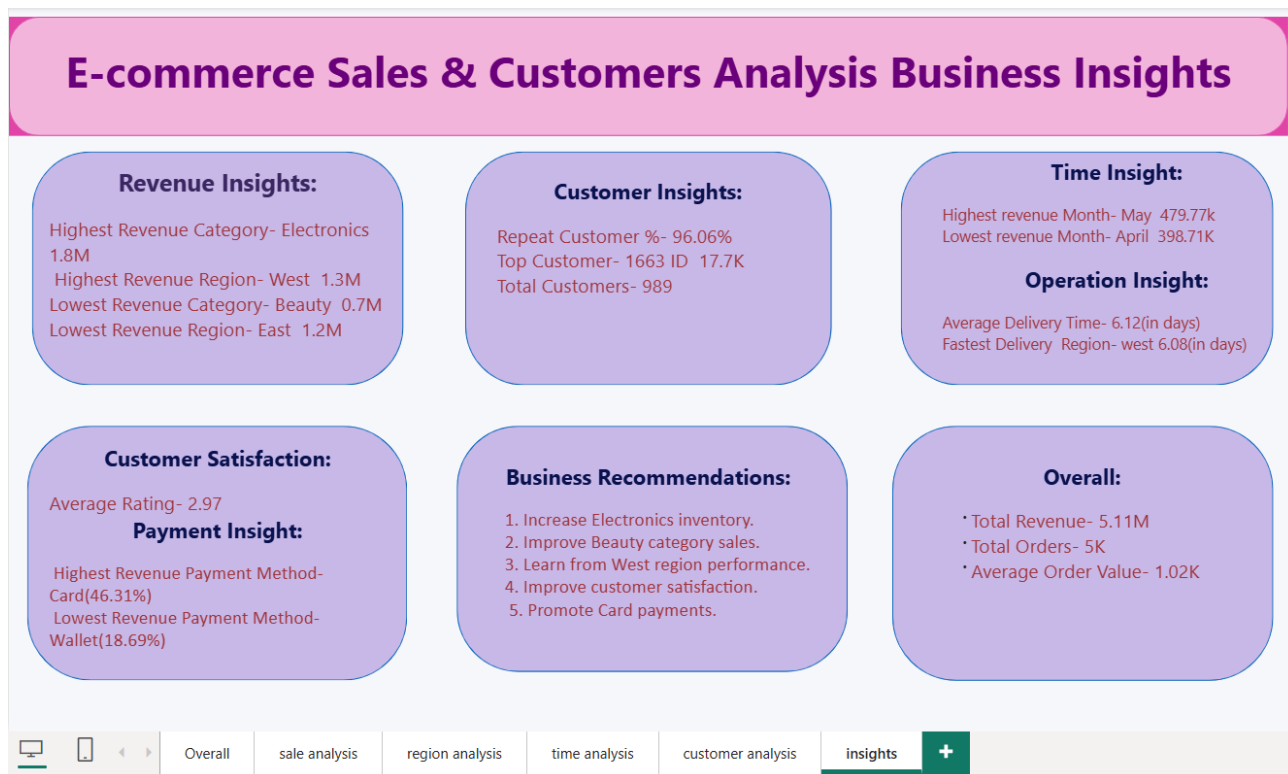


Key Finding:

- 96.06% of customers placed more than one order, indicating a high repeat-order pattern in this dataset.
- Customer 1663 is the highest spender at 17.68K.
- Card is the highest-revenue payment method at 2.37M (46.31%).

Dashboard Page 6 — Business Insights

Executive summary of the key findings, customer behavior, operational performance, payment behavior, and recommended business actions.



9. Consolidated Business Insights

- Overall performance: The business generated approximately 5.11M from 5,000 orders.
- Product performance: Electronics is the leading revenue category at approximately 1.83M.
- Regional performance: West is the highest-revenue region at approximately 1.35M.
- Customer behavior: 96.06% of customers placed more than one order, indicating a high repeat-order pattern in the dataset.
- Payment behavior: Card is the highest-revenue payment method, contributing approximately 46.31% of total revenue.
- Time trend: May is the strongest month while April is the weakest month in the month-of-year view.
- Customer satisfaction: The overall rating is 2.97, which indicates an opportunity to improve the customer experience.
- Operations: West has the fastest average delivery time at approximately 6.08 days.

10. Business Recommendations

- Maintain strong Electronics inventory and identify the products driving its high revenue and quantity.
- Increase Beauty category order frequency through targeted promotions, bundles, and customer campaigns.
- Study West-region practices because it combines the highest revenue with the fastest average delivery.
- Investigate drivers of the 2.97 average customer rating, especially delivery, product quality, returns, and support.
- Continue supporting Card payments while evaluating whether targeted incentives can grow lower-share payment methods.
- Use month-level patterns for inventory and campaign planning, while validating seasonality with year-over-year analysis.

11. Analyst Methodology

- Understand the business questions and define relevant KPIs.
- Inspect the raw data for missing values, inconsistent dates, and schema issues.
- Clean and standardize the data using Python/Pandas.
- Load the cleaned data into SQL for analytical querying.
- Use aggregations, GROUP BY, HAVING, joins, CTEs, and window functions to answer business questions.
- Build Power BI measures and visuals for management-friendly reporting.
- Translate results into business insights rather than only describing charts.
- Convert insights into practical business recommendations.

12. Project Explanation

I worked on an e-commerce sales and customer performance analysis project using order-level data. I first used Python and Pandas to inspect and standardize the data, including the mixed-format order date field. I then used SQL to analyze category, regional, payment, time, and customer performance, including advanced questions using CTEs and window functions. Finally, I built a Power BI dashboard with KPI cards, category and regional analysis, monthly trends, customer analysis, and business insights. The goal was to convert raw order data into business-focused findings and actionable recommendations.

13. Conclusion

This project demonstrates an end-to-end analytics workflow: data preparation in Python, business analysis in SQL, and interactive storytelling in Power BI. The strongest portfolio value comes from connecting each KPI and visual to a business question, explaining the resulting insight, and proposing a practical action.